

Homework Set 8

Due November 7, 2022 @ 8:30am in Gradescope and WileyPlus
(no late submissions for Gradescope given the extended due date)

This assignment is to be completed in your homework groups. Be sure to include the group number as well as first and last names of all contributing group members in addition to the assignment number and instructor name on the group cover sheet (see template in Moodle). Gradescope will ask you to associate the names of all group members when you submit the assignment. **The entire (Gradescope) assignment must be in one person's handwriting.** If you use an equation solving tool (calculator, APEX, Solver, MATLAB), write out the equation(s) and note the tool that you used; otherwise, show all hand calculations in equation solving.

- (15 points)** A vapor mixture of *n*-butane (B) and *n*-hexane (H) contains 50.0 mole% butane at 120°C and 1.0 atm. A stream of this mixture flowing at a rate of 150.0 L/s is cooled and compressed, causing some but not all of the vapor to condense. (Treat this process as a single-unit operation.) Liquid and vapor product streams emerge from the process in equilibrium at T (°C) and 1100 mm Hg. The vapor product contains 60.0 mole% butane.
 - Draw and label a flowchart. Perform a degree-of-freedom analysis to show that you have enough information to determine the required final temperature T , the composition of the liquid product (component mole fractions), and the molar flow rates of the liquid and vapor products from the given information and Antoine expressions for the vapor pressures $p_B^*(T)$ and $p_H^*(T)$. Just identify the equations—for example, mole balance on butane or Raoult's law for hexane—but don't write them yet.
 - Write in order the equations you would use to determine the quantities listed in Part (a) and also the fractional condensation of hexane (mol H condensed/mol H fed). In each equation, circle the variable for which you would solve. Do no algebra or calculations.
 - Complete the calculations either manually or with an equation-solving program.
 - State three assumptions you made that could lead to errors in the calculated quantities.
- (15 points)** Vapor–liquid equilibrium calculations can sometimes be simplified through the use of a quantity called the *relative volatility*, which may be defined in terms of the following depiction of vapor and liquid phases in equilibrium:

Vapor: $y_i, y_j, y_k, \dots$	y_i = mole fraction of species i in vapor
Liquid: $x_i, x_j, x_k, \dots$	x_i = mole fraction of species i in liquid

The relative volatility of species i to species j is

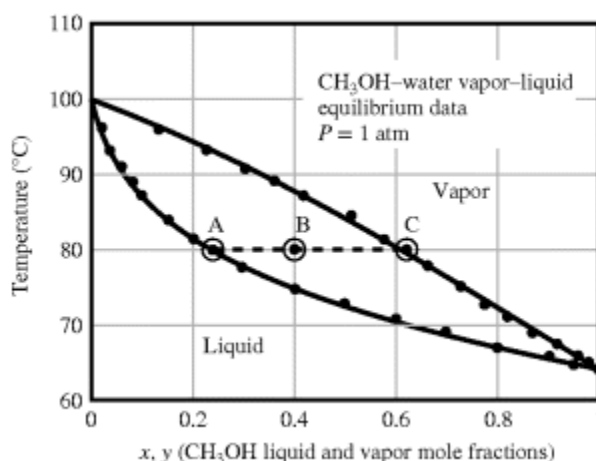
$$\alpha_{ij} = \frac{y_i/x_i}{y_j/x_j} = \frac{y_i/y_j}{x_i/x_j}.$$

If α_{ij} is much greater than 1, species i is much more volatile than species j (i.e., it has a much greater tendency to vaporize at the system temperature and pressure); conversely, if α_{ij} is much less than 1, species i is much less volatile than species j . The closer α_{ij} is to 1, the more difficult it is to separate species i from species j by a process such as distillation or partial condensation of a vapor mixture.

- Show that the relative volatility of species A to species B, α_{AB} , equals the ratio of vapor pressures at the system temperature, p_A^*/p_B^* , if both species obey Raoult's law and follow ideal-gas behavior.
- Determine the relative volatility of styrene to ethylbenzene at 85°C and the relative volatility of benzene to ethylbenzene at the same temperature. Which pair would you classify as more difficult to separate by distillation?
- Show that for a binary mixture of i and j ,

$$y_i = \frac{\alpha_{ij}x_i}{1 + (\alpha_{ij} - 1)x_i}$$

- (d) Apply the equation from Part (c) to a benzene–ethylbenzene system at 85°C, using it to estimate the mole fractions of benzene in the vapor phase in equilibrium with liquids having benzene mole fractions of 0.0, 0.2, 0.4, 0.6, 0.8, and 1.0. Then calculate the total system pressure for each of these six conditions.
3. (15 points) A methanol–water feed stream is introduced to a vaporizer in which a molar fraction f of the feed is vaporized. The feed has a methanol mole fraction of $x_F = 0.4$, and the vaporizer operates at a pressure of 1 atm absolute and 80°C. Vapor and liquid leaving the device are in equilibrium at the temperature and pressure of the system and have methanol mole fractions of y and x , respectively. A Txy diagram for methanol–water mixtures at 1 atm absolute is shown below. The feed to the vaporizer and the liquid and vapor product streams are shown as points B, A, and C, respectively.



- (a) Prove that f can be determined from the equation
- $$f = \frac{\text{moles of vapor}}{\text{moles of liquid}} = \frac{x_F - x}{y - x_F}$$
- Use this result to determine f for the specific conditions cited above ($x_F = 0.4$, $T = 80^\circ\text{C}$).
- (b) Use the Txy diagram to estimate the minimum and maximum temperatures at which the given feed stream could be separated into vapor and liquid fractions at 1 atm. In each case, what fraction of the feed would be vaporized?
- (c) The vapor at C is sent to a condenser operated at constant pressure (1 atm). The liquid and vapor product streams leaving the condenser are in equilibrium and in a ratio of 1 mol vapor/1 mol liquid. Estimate the temperature and compositions of the two streams leaving the condenser.
4. (25 points) In this problem you will use a spreadsheet to create a Txy diagram for the benzene–chloroform system at 1 atm. Once the spreadsheet has been created, it can be used as a template for vapor–liquid equilibrium calculations for other species. The calculations will be based on Raoult’s law (i.e., $y_iP = x_i p_i^*$), although we recognize that this relationship may not produce accurate results for benzene–chloroform mixtures.
- (a) Begin by establishing bounds on the system behavior. Look up the normal boiling points of chloroform and benzene and, without performing any calculations, sketch the expected shape of a Txy diagram for these two species at 1 atm.
- (b) Using APEX or Table B.4, estimate the normal boiling points of the two species and compare them to the results in Part (a).
- (c) Prepare a spreadsheet that has a title row “Txy Diagram for Ideal Binary Solution of Chloroform and Benzene.” In the first cell of Row 2, place the label “ P (mm Hg) =” and in

the adjacent cell enter the system pressure, which for this case is 760. In Row 3 place headings for columns: x_C , x_B , T , p_C^* , p_B^* , P , y_C , y_B , and $y_C + y_B$. Not all of these columns are essential, but when filled they will give a complete picture of the system and a final check of the calculations. Carry out the following procedures in each subsequent row:

- Enter values for the mole fraction of chloroform (the first entry should be 1.000 and the last should be 0.000).
- Calculate the mole fraction of benzene by subtracting the value in the previous cell from 1.000.
- Enter an estimate of the equilibrium temperature that is between the two pure-component boiling points.
- Use APEX or Table B.4 to estimate p_C^* and p_B^* from the estimated temperature.
- Calculate p_C and p_B from Raoult's law.
- Calculate p_C and apply the Goal Seek tool to adjust the value of T until $P = 760$ mmHg.
- Calculate y_C and y_B from the partial pressures and P .
- Sum y_C and y_B to be sure they equal 1.000.
- Once you have completed a row for the first value of x_C , you should be able to copy formulas into subsequent rows. When the calculation has been completed for all rows (i.e., $x_C = 0.0, 0.2, 0.4, 0.5, 0.6, 0.8, 1.0$), draw the T_{xy} diagram.

(d) The following vapor–liquid equilibrium data have been obtained for mixtures of chloroform (C) and benzene (B) at 1 atm.

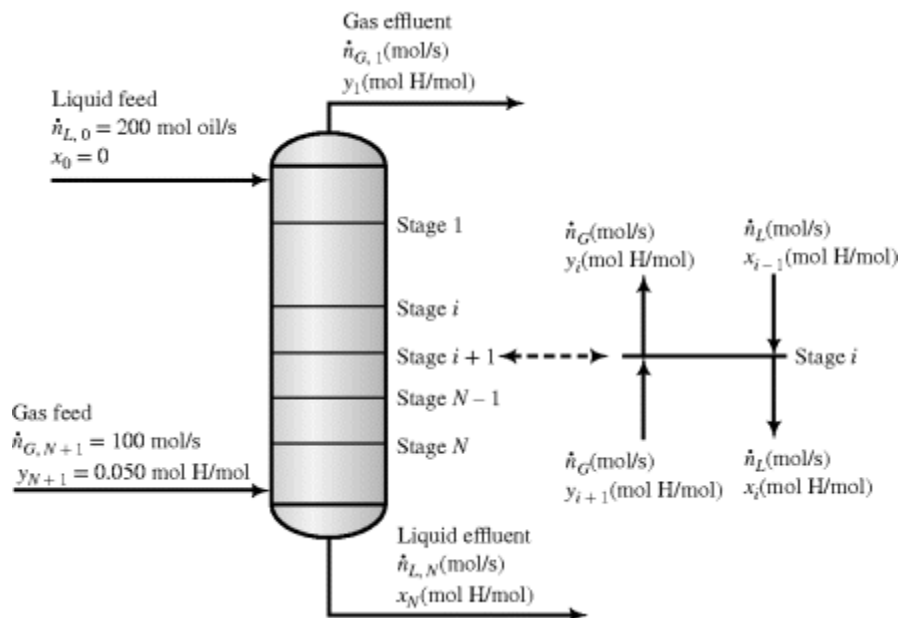
T (°C)	80.6	79.8	79.0	77.3	75.3	71.9	68.9	61.4
x_C	0.00	0.08	0.15	0.29	0.44	0.66	0.79	1.00
y_C	0.00	0.10	0.20	0.40	0.60	0.80	0.90	1.00

Plot these data on the graph generated in Part (c). Estimate the percentage errors in the Raoult's law values of the bubble-point temperature and vapor mole fraction for $x_C = 0.44$, taking the tabulated values to be correct.

(30 points) The remainder of the assignment (4 problems) will be completed individually online using WileyPLUS. Each person will complete and submit their own WileyPLUS problems. *You can work with your team members to solve these problems, although note that each person may have unique values for some variables.* You do not have to turn in any paperwork with this portion of the assignment. The due date for the WileyPLUS completion is the same as for the homework assignment. Note that the WileyPLUS assignment cannot be submitted late.

Honors Contract Problem

The following diagram shows a staged absorption column in which n-hexane (H) is absorbed from a gas into a heavy oil.



A gas feed stream containing 5.0 mole% hexane vapor and the balance nitrogen enters at the bottom of an absorption column at a basis rate of 100 mol/s, and a nonvolatile oil enters the top of the column in a ratio 2 mol oil fed/mol gas fed. The absorber consists of a series of ideal stages (see Problem 6.66), arranged so that gas flows upward and liquid flows downward. The liquid and gas streams leaving each stage are in equilibrium with each other (by the definition of an ideal stage), with compositions related by Raoult's law. The absorber operates at an approximately constant temperature T ($^{\circ}\text{C}$) and 760 mm Hg. Of the hexane entering the column, 99.5% is absorbed and leaves in the liquid column effluent. At the given conditions it may be assumed that N_2 is insoluble in the oil and that none of the oil vaporizes.

- Calculate the molar flow rates and mole fractions of hexane in the gas and liquid streams leaving the column. Then calculate the average values of the liquid and gas molar flow rates in the column, $\bar{n}_L$ (mol/s) and $\bar{n}_G$ (mol/s). For simplicity, in subsequent calculations use the average values for liquid and gas molar flow rates within the column, but the actual values for the corresponding flow rates entering and leaving the column.
- Considering the bottom stage to be ideal, estimate the mole fractions of hexane in the gas leaving that stage (y_N) and in the liquid entering it (x_{N-1}) if the column temperature is 50°C .
- Suppose that x_i and y_i are the mole fractions of hexane in the liquid and gas streams leaving stage i . Derive the following equations from an equilibrium relationship and a mass balance around a section of the column encompassing stage i and the bottom of the column:

$$y_i = \frac{x_i p_i^*(T)}{P} \quad (1)$$

$$x_{i-1} = \frac{x_N \bar{n}_{L,N} + y_i \bar{n}_G - y_{N+1} \bar{n}_{G,N+1}}{\bar{n}_L} \quad (2)$$

Verify that these equations yield the answers you calculated in Part (b).

- Examine the effect of operating temperature on the column by estimating the number of ideal stages necessary to achieve the desired separation. In the calculations, which will be done using a

spreadsheet, take the pressure in the column to be constant at 760 torr, but consider three different operating temperatures: 30°C, 50°C, and 70°C. The calculations will follow a stage-to-stage strategy beginning at the bottom of the column and repeatedly applying (Equations 1) and (2) until the mole fraction of hexane in the vapor leaving the column is less than or equal to that calculated in Part (a). You may use APEX or the Antoine equation and Table B.4 to estimate the hexane vapor pressure. The calculations for the case of $T = 30^\circ\text{C}$ illustrate how to proceed; for this case, $y_{N-1} < y_1 = 0.00263$ after only two stages.

Hexane Absorption								
P (torr)	760	y_{N+1}	0.05	x_0	0	$n_{G,N+1}$ (mol/s)	100	
Fraction removal	0.95	y_1	0.00263	x_N		$n_{L,0}$ (mol/s)	200	
						n_G (mol/s)		
						n_L (mol/s)		
T (degC)	30		T (degC)	50		T (degC)	70	
p^* (torr)	187.1		p^* (torr)			p^* (torr)		
i	$x_i \cdot 10^3$	$y_i \cdot 10^3$	i	$x_i \cdot 10^3$	$y_i \cdot 10^3$	i	$x_i \cdot 10^3$	$y_i \cdot 10^3$
N + 1		50.00			50.00			50.00
N	24.27	5.976						
N - 1	2.754	0.678						

When you set up your spreadsheet, you should be able to accomplish many of your calculations simply by copying cells from one location to another. In any case, *do not go beyond 25 stages* for any temperature, whether or not you achieve the required separation.

- (e) You can see that the number of stages required increases as the column temperature increases. In fact, there is a maximum temperature beyond which the required separation cannot be achieved. At that temperature, the entering gas and leaving liquid are approximately in equilibrium, so that $x_N p^*(T) = y_{N+1} P$. Use either APEX or the Antoine equation to estimate the maximum temperature at which the separation can be achieved.